

mathematical errors in the Distance test. These types of errors show a more fundamental misunderstanding of spatial reasoning by these LLMs, rather than simply a counting problem.

This degradation emphasizes the fact that these models are built for linguistic tasks, and, while presenting some emergent behavior [16] [17], are still limited by their architecture. The models possess the reasoning capabilities to handle small examples of a task, but lose track of the grid and task once scaled.

The differences between the models themselves are just as interesting. It is immediately evident across the tests that GPT-4o is the weakest at spatial reasoning, followed by GPT-4.1. Both Anthropic models consistently outperformed the OpenAI models, with gaps of up to 30% in accuracy. GPT-4o also often failed to provide a parseable response despite establishing extensive regex and prompt specifications, whereas none of the other models encountered any such problems.

The error types are interesting as well. The OpenAI models are less internally consistent than the Anthropic ones, displaying weaker consistency in the Quadrant and Transformation tests. In addition, the OpenAI models made more errors in the Distance test, both in position and computation. However, the OpenAI models were less likely to hallucinate a correct answer in the Word Search test. This behavior, of the OpenAI models performing worse in the computational tests, but better in the language-based one, is likely a result of the models’ different architectures and training.

6 Conclusion

While the Anthropic models outperformed the OpenAI models using this test suite, it is clear that none of the models are particularly good at spatial reasoning. The abilities of the Anthropic models hold promise, but they still deteriorate quickly with scale. We propose a few avenues for future research on this deterioration. For the top models currently released, presentation format and context can significantly affect response quality. For instance, the models tested are all multimodal, and would likely perform better on similar tests presented in image form; future work could focus on directly testing these various modes of input. Other future work could follow up on the changes in grid representation attempted, with an interest in improving efficiency or effectiveness of representations of spatial data in a text-heavy format, in efforts to reduce or minimize the loss in accuracy shown. Further work could also focus on improving the counting and mathematical capabilities of LLMs, as these two skills were the root of many of the inaccurate responses. Finally, research could focus on other methods of output, such as code generation or full boards.

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